

DAY 1

Introduction to Financial Modeling

FMVA Certification Prep: Financial Modeling and Valuation Bootcamp

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This chapter builds working competence in Introduction to Financial Modeling: the concepts, the standard procedure, and the judgment to apply it under exam conditions.

Concepts and vocabulary

Financial modeling sits at the intersection of accounting, finance, and applied mathematics. Before building any model, you need a shared vocabulary — a precise set of terms that lets analysts, managers, and auditors speak the same language and audit each other's work without ambiguity.

Core Definitions

A financial model is a structured, quantitative representation of a business situation, transaction, or investment scenario. Its purpose is not necessarily to mirror reality perfectly — scholars in the philosophy of financial modeling have argued that a model's primary task is to demonstrate the possibility of a transaction under a defined set of assumptions, rather than to replicate an ongoing real-world situation exactly.

Several foundational terms appear in virtually every model:

- **Input (assumption):** An independently chosen value that drives the model — for example, a revenue growth rate or a discount rate. Inputs are the levers an analyst adjusts during sensitivity analysis.
- **Output:** A calculated result produced by the model — net income, enterprise value, or free cash flow.
- **Driver:** A specific input whose change has an outsized effect on a key output. Identifying drivers early is one of the first tasks in model design.
- **Hardcode (constant):** A number typed directly into a cell or formula rather than referenced from an assumption block. Best practice minimizes hardcodes.
- **Link:** A reference from one cell or worksheet to another, creating a traceable chain from inputs to outputs.
- **Scenario:** A complete, internally consistent set of inputs representing one possible future state (e.g., base case, upside, downside).
- **Sensitivity analysis:** A systematic test of how outputs change when one or more inputs vary, holding others constant.

Where Financial Modeling Fits in Practice

Financial models appear across the full spectrum of corporate finance work: budgeting and forecasting, merger and acquisition valuation, leveraged buyout analysis, project finance, and equity research. In each context, the model translates financial statements — which report historical position and performance — into forward-looking projections that support economic decisions. Because financial statements are designed to be understandable, relevant, reliable, and comparable, a well-built model preserves those qualities in its outputs.

Notation Conventions

Standard Financial Modeling Procedure



Fixed sequence transforms raw assumptions into actionable financial insights

Consistent notation prevents errors and speeds review. The most widely adopted conventions include:

| Convention | Meaning |

| ---|---|

| Blue font | Hardcoded input |

| Black font | Formula referencing other cells |

| Green font | Link to another worksheet or file |

| t | Time period index (e.g., Year 1 = t = 1) |

| r | Discount rate or required return |

| g | Growth rate |

| CF_t | Cash flow in period t |

A present-value formula using this notation looks like:

,

$$PV = CF_t / (1 + r)^t$$

,

Step-by-Step Procedure: Mapping Vocabulary to a New Model

- Identify the decision the model must support (e.g., "Should we acquire this company?").
- List all inputs and separate them into an assumption block at the top of the model.
- Label each driver — the two or three inputs most likely to swing the output materially.

- Build output formulas that reference only the assumption block, never hardcoded numbers buried in calculations.
- Define at least three scenarios (base, upside, downside) by varying the drivers.
- Run sensitivity analysis on the top two drivers to quantify output ranges.

Worked Example: Terminology Applied to a Mini Scenario

A small retailer expects \$500,000 in revenue in Year 1, growing at 8 % per year for three years. The analyst wants to project Year 3 revenue.

Step 1 — Identify the input and driver.

Revenue in Year 1 (\$500,000) and growth rate (8 %) are both inputs; the growth rate is the primary driver because small changes in it move Year 3 revenue significantly.

Step 2 — Write the formula.

,

$\text{Revenue}_t = \text{Revenue}_0 \times (1 + g)^t$

,

Step 3 — Calculate Year 3 output.

,

$\text{Revenue}_3 = \$500,000 \times (1 + 0.08)^3$

$= \$500,000 \times 1.2597$

$= \$629,856$

,

Step 4 — Quick sensitivity check (driver varies ± 2 %).

| Growth Rate | Year 3 Revenue |

|---|---|

| 6 % | \$595,508 |

| 8 % (base) | \$629,856 |

| 10 % | \$665,500 |

The \$70,000 spread across scenarios illustrates why the growth rate is labeled the driver.

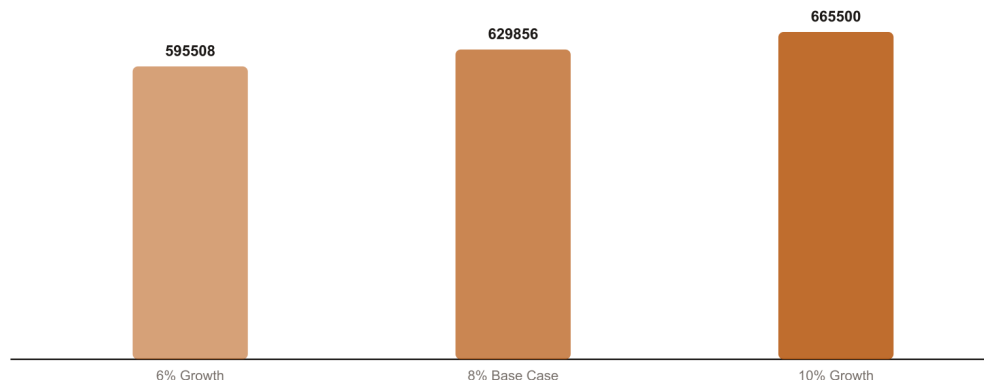
Common mistake: Hardcoding the growth rate (typing 1.08 directly into the revenue formula) instead of referencing a dedicated input cell. When you later want to run the sensitivity table above, you must hunt through every formula to change the rate — a slow, error-prone process that a single referenced input cell eliminates entirely.

The standard procedure

Financial modeling follows a disciplined, repeatable procedure that transforms raw assumptions into actionable insights. Mastering this procedure—inputs, steps, outputs, and checks—is the foundation of professional modeling practice.

Inputs

Revenue Sensitivity: Growth Rate Impact



\$70K spread across scenarios proves growth rate is the primary model driver

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Every model begins with clearly defined inputs. These fall into two categories:

- Hard-coded assumptions — values you type directly into the model, such as a revenue growth rate, tax rate, or capital expenditure figure. These should be clearly labeled and ideally isolated in a dedicated assumptions section.
- Linked data — historical financial statements, market prices, or third-party data that feed into the model dynamically.

Good practice demands that inputs are never buried inside formulas. A reader—or your future self—must be able to locate every assumption instantly. Color-coding inputs (commonly blue font) is a widely adopted convention.

Steps

Once inputs are assembled, the standard procedure unfolds in a fixed sequence:

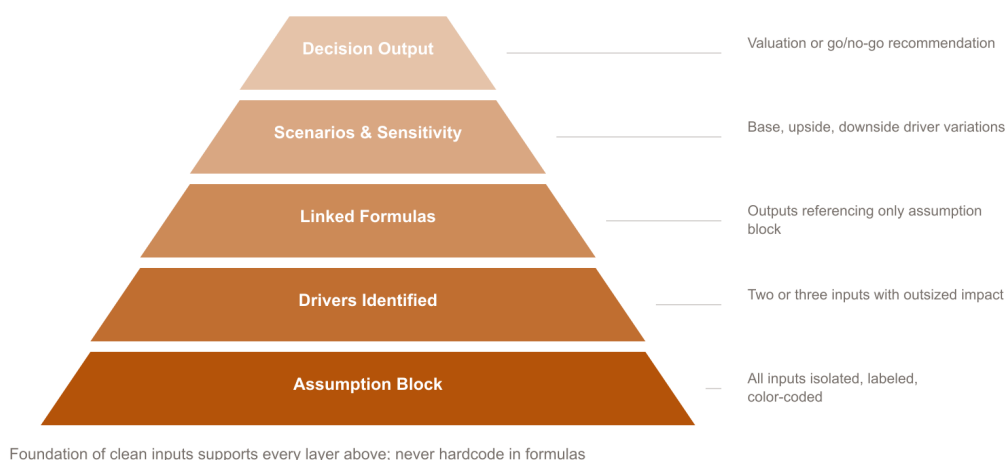
- Define the scope and time horizon. Decide whether you are modeling quarterly or annual periods, and how many periods forward.
- Build the income statement. Start with revenue, apply growth rates, subtract costs, and arrive at operating income (EBIT), then net income.
- Build the balance sheet. Project assets, liabilities, and equity using drivers tied to the income statement (e.g., days sales outstanding for receivables).
- Build the cash flow statement. Derive operating, investing, and financing cash flows—ideally by linking directly to the income statement and balance sheet rather than entering values manually.
- Apply the valuation or decision layer. This might be a discounted cash flow (DCF) analysis, a debt-service coverage calculation, or a scenario comparison.
- Run scenarios and sensitivities. Vary key inputs (growth rate, margin, discount rate) to understand how outputs respond.

Outputs

Outputs are the model's deliverables: projected financial statements, a valuation range, an IRR, a debt schedule, or a summary dashboard. Outputs must be clearly separated from inputs and should never be manually overwritten—they must always trace back to a formula chain rooted in the inputs.

Checks

Financial Model Component Hierarchy



Foundation of clean inputs supports every layer above; never hardcode in formulas

A model without checks is a liability. Standard checks include:

- Balance sheet balance: Total assets must equal total liabilities plus equity in every period.
- Cash flow reconciliation: The closing cash balance on the cash flow statement must match the cash line on the balance sheet.
- Circular reference flags: If the model uses iterative calculations (e.g., interest on a revolver), confirm that iteration is intentional and controlled.

Many modelers add a dedicated "checks" row that displays 0 when everything balances and flags an error otherwise.

Fully Worked Numerical Example

Scenario: Project Year 1 net income and ending cash for a small retailer.

Inputs:

- Base revenue: \$500,000
- Revenue growth rate: 10%
- Cost of goods sold (COGS): 60% of revenue
- Operating expenses (OpEx): \$80,000 (fixed)

- Tax rate: 25%
- Beginning cash: \$40,000
- Capital expenditure (CapEx): \$15,000
- Depreciation: \$10,000

Step 1 — Revenue:

$$\text{Revenue} = \$500,000 \times (1 + 0.10) = \$550,000$$

Step 2 — Gross profit:

$$\text{Gross Profit} = \$550,000 - (\$550,000 \times 0.60) = \$550,000 - \$330,000 = \$220,000$$

Step 3 — EBIT:

$$\text{EBIT} = \$220,000 + \$80,000 = \$300,000$$

Step 4 — Net income:

$$\text{Net Income} = \$300,000 \times (1 - 0.25) = \$225,000$$

Step 5 — Operating cash flow (indirect method):

$$\text{Operating Cash Flow} = \text{Net Income} + \text{Depreciation} = \$225,000 + \$10,000 = \$235,000$$

Step 6 — Ending cash:

$$\text{Ending Cash} = \$40,000 + \$235,000 - \$15,000 = \$255,000$$

Check: Confirm that the \$255,000 ending cash will appear identically on both the cash flow statement and the balance sheet cash line. If they differ, a formula link is broken.

Common mistake: Modelers often hard-code outputs—typing \$105,000 directly into the net income cell instead of building the formula chain. This severs the link between inputs and outputs, meaning the model will not update when assumptions change, which defeats its entire purpose.

The power of this procedure lies in its repeatability. Once you can walk through a clean numerical pass from inputs to checks, you can scale the same logic to multi-year, multi-entity models with confidence.

Applications and pitfalls

Applications and Pitfalls

Financial modeling is most valuable when it connects a real business question to a structured, auditable set of assumptions and calculations. Understanding where models are typically deployed—and where they most often go wrong—is essential preparation for both professional practice and

certification exams.

Typical Use Cases

Financial models appear across nearly every corner of corporate finance and investment analysis:

- Valuation — Discounted cash flow (DCF) models estimate intrinsic value by projecting free cash flows and discounting them at an appropriate rate.
- Budgeting and forecasting — Operating models translate revenue drivers and cost assumptions into projected income statements, balance sheets, and cash flow statements.
- Merger and acquisition analysis — Merger models assess accretion or dilution to earnings per share and evaluate deal pricing.
- Scenario and sensitivity analysis — Models stress-test outcomes by varying key inputs (growth rate, margins, discount rate) to understand the range of possible results.
- Credit analysis — Lenders use models to assess debt service coverage and covenant compliance under different economic conditions.

In each case, the model's purpose is not to predict the future with certainty but to demonstrate the plausibility of a scenario given a defined set of assumptions—a point increasingly emphasized in the philosophy of financial modeling.

Common Mistakes

Even experienced analysts make systematic errors. The most damaging tend to fall into three categories:

- Hardcoding numbers inside formulas instead of linking to a dedicated assumptions section, making the model opaque and difficult to audit.
- Circular references left unresolved, which cause Excel to iterate unpredictably or return errors.
- Mixing time periods, such as using an annual discount rate against monthly cash flows without adjustment.
- Ignoring the terminal value's dominance in a DCF—in many models, the terminal value represents 60–80 % of total value, so a small error in the perpetuity growth rate has an outsized effect.
- Forgetting to sanity-check outputs against market comparables or simple rules of thumb.

Sanity Checks

Before presenting any model, run these quick tests:

- Does the balance sheet balance ($\text{Assets} = \text{Liabilities} + \text{Equity}$)?
- Does net income flow correctly into retained earnings?
- Do the implied valuation multiples (e.g., EV/EBITDA) fall within a reasonable industry range?
- Does a 1 % change in the discount rate produce a directionally sensible change in value?
- Is the terminal growth rate below the long-run nominal GDP growth rate?

Worked Example: A Flawed DCF Corrected Step by Step

Scenario: An analyst values a small retailer projecting \$500,000 in free cash flow (FCF) in Year 1, growing at 4 % per year in perpetuity, with a WACC of 9 %.

Step 1 — The flawed attempt.

The analyst calculates the terminal value using the Gordon Growth Model but accidentally uses the current-year FCF rather than next-year FCF:

$$\text{Terminal Value} = \text{FCF } \text{€} \div (\text{WACC} - g)$$

$$\text{Terminal Value} = \$500,000 \div (0.09 - 0.04) = \$10,000,000$$

This appears reasonable at first glance, but it is wrong.

Step 2 — Identify the error.

The Gordon Growth Model requires the cash flow one period ahead of the valuation date. Using FCF € (the current year) understates the terminal value because it ignores one year of growth.

Step 3 — Apply the correct formula.

$$\text{Terminal Value} = \text{FCF} \bullet \times (1 + g) \div (\text{WACC} - g)$$

Wait—if FCF • is already Year 1's cash flow, the terminal value at the end of the explicit forecast period uses FCF in the first year of the perpetuity, which is Year 2:

$$\text{FCF} \bullet = \$500,000 \times (1.04) = \$520,000$$

$$\text{Terminal Value} = \$520,000 \div (0.09 - 0.04) = \$10,400,000$$

Step 4 — Discount the terminal value to present.

Assuming the terminal value is received at the end of Year 1:

$$\text{PV of Terminal Value} = \$10,400,000 \div (1.09)^1 = \$9,541,284$$

Step 5 — Add the PV of Year 1 FCF.

$$\text{PV of FCF} \bullet = \$500,000 \div 1.09 = \$458,716$$

$$\text{Total Value} = \$458,716 + \$9,541,284 = \$10,000,000$$

Step 6 — Sanity check. Divide total value by FCF • to get an implied Price/FCF multiple of 20×, consistent with a mature retailer at a 9 % discount rate. The result is plausible.

Common mistake: Using the current-period cash flow (FCF €) in the terminal value formula instead of the next-period cash flow (FCF •) is one of the most frequent DCF errors on exams and in practice. Always grow the final projected cash flow by $(1 + g)$ before dividing by $(\text{WACC} - g)$.

Catching errors like this one—through careful formula construction and systematic sanity checks—is what separates a reliable model from a misleading one.

